

ENGINEERING CHANGE ORDER — CAPITAL MODIFICATION PROJECT

Manufacturer Inc. • Manufacturing Division • Plant Engineering

ECO Number	ECO-2025-017
Related Asset	Custom Milling Machine (register ID 1002)
Base Equipment	Hartwig-Kessler HK-880 5-Axis Milling Center, Serial HK880-9F1204
Project Raised	October 4, 2024
Work Completed	December 19, 2024
Final Acceptance / Placed in Service	December 28, 2024 (acceptance) • January 1, 2025 (in service)
Approval Route	Divisional Controller and VP Operations (per FIN-POL policy approval matrix)
Status	Closed — costs transferred out of construction in progress

1. Business Justification

The Division's Tier-2 housing program requires simultaneous five-axis contouring at tolerances the OEM machine cannot hold in its standard configuration. Rather than procure a purpose-built machine, Plant Engineering elected to modify the existing HK-880 base machine using the Division's own spindle and control design. The resulting configuration is specific to the Division's process and is not offered by Hartwig-Kessler as a catalogue option.

2. Scope of Work Performed

- 2.1. Removal of the OEM 4-axis control head and substitution of the Division's proprietary five-axis spindle assembly (drawing set MD-5AX-2024, company-owned intellectual property).
- 2.2. Replacement of the OEM tool-changer interface with a bespoke 40-station carousel keyed to the Division's tooling standard. The original interface plate was machined away and cannot be reinstated.
- 2.3. Replacement of the OEM CNC controller firmware with Division-developed motion control firmware embodying the Division's proprietary toolpath compensation algorithms. The OEM firmware image cannot be reloaded without replacing the substituted control hardware.
- 2.4. Structural reinforcement of the machine bed and re-pouring of the epoxy granite base to absorb the additional dynamic load of the substituted spindle.
- 2.5. Custom calibration and process qualification against the Division's internal acceptance specification MD-QA-118.

3. Cost Summary

Line	Description	Amount
1	Base machine — Hartwig-Kessler HK-880, purchase price net of trade discount	\$96,400.00
2	Freight, rigging and site preparation	\$5,600.00
3	Proprietary five-axis spindle assembly (fabrication and materials)	\$19,850.00
4	Bespoke tool-changer carousel and interface plate	\$7,300.00
5	Control hardware substitution and firmware integration labor	\$6,420.00
6	Bed reinforcement and epoxy granite re-pour	\$2,180.00

Line	Description	Amount
7	Custom calibration, process qualification and acceptance testing	\$2,250.00
	Total capitalized cost transferred to fixed asset register ID 1002	\$140,000.00

3.1. Lines 3 through 7 were accumulated in construction in progress during Q4 2024 and transferred to the fixed asset register together with lines 1 and 2 on final acceptance. The modified machine is carried as a single asset record; no component-level cost records are maintained.

3.2. Operator training of \$4,100.00 and the three-year extended service contract of \$6,750.00 were expensed as incurred and are not included in the total above.

4. Engineering Notes and Post-Modification Status

4.1. Hartwig-Kessler has confirmed in writing that the substitution of the control head and spindle assembly voids the standard equipment warranty in its entirety, effective on the date of substitution.

4.2. The machine cannot be returned to its manufacturer-standard configuration without disassembly and replacement of the substituted spindle, tool-changer interface plate, control hardware and reinforced bed. Plant Engineering assesses the cost of such a reversal as exceeding the residual value of the machine.

4.3. The used-equipment brokers canvassed by Procurement (Meridian Machinery, Baltic Surplus) both declined to quote a resale value on the machine in its modified configuration, citing the absence of a secondary market for a non-standard control architecture. No trade-in or buy-back arrangement is available from the OEM.

4.4. The machine is dedicated to the Tier-2 housing program and cannot be redeployed to other work cells without a further modification project of comparable scope.

4.5. The Division's proprietary designs and firmware embodied in this machine are subject to the confidentiality controls in the Division's IP register; the machine may not be sold, leased or transferred to a third party in its modified state without IP counsel review.

5. Sign-off

Role	Name	Signature	Date
Plant Engineering Manager	R. Vasquez	/s/ R. Vasquez	Dec 19, 2024
Quality Assurance Lead	T. Okonkwo	/s/ T. Okonkwo	Dec 27, 2024
VP Operations	M. Oyelaran	/s/ M. Oyelaran	Dec 28, 2024
Divisional Controller	H. Lindqvist	/s/ H. Lindqvist	Dec 30, 2024